

7.7 Practice - Solving Rational Equations

Solve the following equations for the given variable:

1) $3x - \frac{1}{2} - \frac{1}{x} = 0$

2) $x + 1 = \frac{4}{x+1}$

3) $x + \frac{20}{x-4} = \frac{5x}{x-4} - 2$

4) $\frac{x^2+6}{x-1} + \frac{x-2}{x-1} = 2x$

5) $x + \frac{6}{x-3} = \frac{2x}{x-3}$

6) $\frac{x-4}{x-1} = \frac{12}{3-x} + 1$

7) $\frac{2x}{3x-4} = \frac{4x+5}{6x-1} - \frac{3}{3x-4}$

8) $\frac{6x+5}{2x^2-2x} - \frac{2}{1-x^2} = \frac{3x}{x^2-1}$

9) $\frac{3m}{2m-5} - \frac{7}{3m+1} = \frac{3}{2}$

10) $\frac{4x}{2x-6} - \frac{4}{5x-15} = \frac{1}{2}$

11) $\frac{4-x}{1-x} = \frac{12}{3-x}$

12) $\frac{7}{3-x} + \frac{1}{2} = \frac{3}{4-x}$

13) $\frac{7}{y-3} - \frac{1}{2} = \frac{y-2}{y-4}$

14) $\frac{2}{3-x} - \frac{6}{8-x} = 1$

15) $\frac{1}{x+2} - \frac{1}{2-x} = \frac{3x+8}{x^2-4}$

16) $\frac{x+2}{3x-1} - \frac{1}{x} = \frac{3x-3}{3x^2-x}$

17) $\frac{x+1}{x-1} - \frac{x-1}{x+1} = \frac{5}{6}$

18) $\frac{x-1}{x-3} + \frac{x+2}{x+3} = \frac{3}{4}$

19) $\frac{3}{2x+1} + \frac{2x+1}{1-2x} = 1 - \frac{8x^2}{4x^2-1}$

20) $\frac{3x-5}{5x-5} + \frac{5x-1}{7x-7} - \frac{x-4}{1-x} = 2$

21) $\frac{x-2}{x+3} - \frac{1}{x-2} = \frac{1}{x^2+x-6}$

22) $\frac{x-1}{x-2} + \frac{x+4}{2x+1} = \frac{1}{2x^2-3x-2}$

23) $\frac{3}{x+2} + \frac{x-1}{x+5} = \frac{5x+20}{6x+24}$

24) $\frac{x}{x+3} - \frac{4}{x-2} = \frac{-5x^2}{x^2+x-6}$

25) $\frac{x}{x-1} - \frac{2}{x+1} = \frac{4x^2}{x^2-1}$

26) $\frac{2x}{x+2} + \frac{2}{x-4} = \frac{3x}{x^2-2x-8}$

27) $\frac{2x}{x+1} - \frac{3}{x+5} = \frac{-8x^2}{x^2+6x+5}$

28) $\frac{x}{x+1} - \frac{3}{x+3} = \frac{-2x^2}{x^2+4x+3}$

29) $\frac{x-5}{x-9} + \frac{x+3}{x-3} = \frac{-4x^2}{x^2-12x+27}$

30) $\frac{x-3}{x+6} + \frac{x-2}{x-3} = \frac{x^2}{x^2+3x-18}$

31) $\frac{x-3}{x-6} + \frac{x+5}{x+3} = \frac{-2x^2}{x^2-3x-18}$

32) $\frac{x+3}{x-2} + \frac{x-2}{x+1} = \frac{9x^2}{x^2-x-2}$

33) $\frac{4x+1}{x+3} + \frac{5x-3}{x-1} = \frac{8x^2}{x^2+2x-3}$

34) $\frac{3x-1}{x+6} - \frac{2x-3}{x-3} = \frac{-3x^2}{x^2+3x-18}$